

ZKTrustLLM-Agents L4 Supervisor Milestone Report: Automation, Audit, IPFS, and On-Chain Validation

Created: 2026-05-09T18:01:00.237719+00:00

1. Purpose

This supervisor milestone report documents the ZKTrustLLM-Agents Level 4 automation chain from Step 96 to Step 105. It connects agentic experiment automation, KPI-driven decision making, policy-gated remediation, governance, hash-chained audit evidence, IPFS anchoring, and local on-chain validation.

2. Full Process Architecture

Layer	Implemented Steps	Role
Automation	Step 96	Runs L4 experiments and validates outputs.
Decision	Step 97	Reads KPI artifacts and recommends next action.
Remediation	Step 98	Executes approved actions after human approval.
Governance	Steps 99-100	Defines policy gates, approval boundaries, and rollback notes.
Audit	Step 101	Creates hash-chained evidence over automation artifacts.
Anchor	Step 102	Creates IPFS and blockchain-ready commitment evidence.
Trust Plane	Steps 103-104	Stores anchor on local Hardhat and validates replay rejection.
Policy-Gated Chain	Step 105	Runs on-chain validation through approval-gated automation.

3. Step-by-Step Progress

Step	Component	Status / Decision	Evidence artifact
Step 96	Agentic automation suite	PASS	results/l4_agentic_automation/agentic_automation_summary.json
Step 97	Closed-loop KPI decision agent	GENERATE_SUPERVISOR_REPORT	results/l4_closed_loop_agent/closed_loop_kpi_decision.json
Step 98	Closed-loop remediation executor	EXECUTED_PASS	results/l4_remediation_executor/remediation_execution_plan.json
Step 99	Automation governance dashboard	JSON_PRESENT	results/l4_automation_governance/automation_governance_dashboard.json
Step 100	Policy-gated autonomous scheduler	EXECUTED_PASS	results/l4_policy_scheduler/policy_scheduler_summary.json
Step 101	Hash-chained automation audit ledger	FINAL_HASH_0db58b3f2c09	results/l4_automation_audit_ledger/automation_audit_ledger.json
Step 102	IPFS/blockchain-ready audit anchor	IPFS_ADD_PASS	results/l4_audit_anchor/audit_ledger_anchor_record.json
Step 103	On-chain audit-anchor registry	ANCHOR_ID_0	results/l4_onchain_anchor/onchain_audit_anchor_result.json
Step 104	On-chain negative-security validation	PASS	results/l4_onchain_anchor_negative/onchain_anchor_negative_security_result.json
Step 105	Policy-gated on-chain validation	EXECUTED_PASS	results/l4_policy_onchain_validation/policy_onchain_validation_summary.json

4. Key Evidence Values

Evidence	Value
step96OverallStatus	PASS
step97PrimaryAction	GENERATE_SUPERVISOR_REPORT
step98Status	EXECUTED_PASS
step100SchedulerStatus	EXECUTED_PASS
step101EntryCount	14
step101FinalLedgerHash	0db58b3f2c09e530c280dc6cd795a53df54e3337b4a46aa9c70fe6ea166c2567
step102IpfsStatus	IPFS_ADD_PASS
step102IpfsCid	QmW7VeYs3fP5kk92JstT8tAEkQytgseH4SfhKRpcSwU112
step102CommitmentHash	50aa798c5968470f172f5e91cb9d80e25b6ac018d3acde55a424f32ec1c2c8ec
step103GasUsed	320095
step103TxHash	0x3aa06732baa252f303eb24a6b20c118a6a3c665b7e8cbcb6dce0a2b9d3053277
step104OverallStatus	PASS
step105OverallStatus	EXECUTED_PASS

5. Scientific Contribution

- The project now demonstrates a Level 4 agentic automation workflow rather than independent scripts.
- The automation chain is KPI-aware, policy-gated, approval-gated, and supervisor-auditable.
- The audit ledger gives the automation process a tamper-evident evidence chain.
- The IPFS and blockchain-ready anchor connects the automation layer to the ZKTrustLLM evidence/trust-plane design.
- The local Hardhat registry and negative-security tests show replay-resistant anchor validation.

6. Governance Boundary

The automation does not automatically push code, modify Git history, submit papers, email supervisors, deploy public-chain contracts, spend real funds, or delete evidence. Sensitive actions remain explicitly human-approved and policy-gated.

7. Recommended Next Steps

Next Step	Purpose
Step 107	Create journal-ready architecture figures and result tables for Steps 96-105.
Step 108	Add repeated-run statistics and confidence intervals for automation and on-chain validation.
Step 109	Move from ephemeral Hardhat runs to a persistent local node or controlled testnet dry-run.
Step 110	Draft journal methodology and evaluation sections around policy-gated agentic automation.